

An Internship Report
Of
C++ Programming

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B-Tech (CSE)



By

Roll Number of Student:- 25SCS1003005141

Name of Student: Sukrit Kumar

Batch: 2025-29

CANDIDATE'S DECLARATION

I, SUKRIT KUMAR, do hereby declare that the internship report titled “**C++ PROGRAMMING**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:- Sukrit Kumar

Student Name: **SUKRIT KUMAR**

Roll No.: **25SCS1003005141**

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the **THIRANEX VIRTUAL INTERNSHIP** for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of **C++ PROGRAMMING**. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature: Sukrit Kumar

Student Name: **SUKRIT KUMAR**

Roll No.: **25SCS1003005141**

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3. INTERNSHIP COMPLETION CERTIFICATE



Internship Offer Letter



Date: 10 August 2026

TO,

Sukrit Kumar

Intern ID: THX-AUG1026-450

Subject: Offer Letter for Internship in C++ Programming

Dear **Sukrit Kumar**,

Following our selection process, we are pleased to offer you an internship position at **Thiranex**. We believe your skills and enthusiasm will be a great addition to our team.

Internship Role	Intern - C++ Programming
Commencement Date	10 Aug 2026
Completion Date	09 Sept 2026
Work Mode	Remote / Project-Based

During this tenure, you will work on practical projects under industry mentorship. Your progress will be reviewed periodically, and your performance will determine the successful completion of the internship.

We look forward to a mutually beneficial learning journey. Please confirm your acceptance by starting your onboarding tasks.



This is an electronically generated document. No physical signature is required.
For verification, contact Thiranex Verification Cell.
www.thiranex.in



Thiranex

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COMPLETED

Due Date: Immediate Submission

Submitted

Mandatory: Offer Letter & LinkedIn Post

Download your offer letter, post it on LinkedIn tagging Thiranex, and share the post URL here.

Completed Successfully!

1

Due Date: 17 Aug 2026

Submitted on track

Student Management System

To develop a console-based Student Management System in C++ that efficiently manages student records using file handling and menu-driven operations.

Watch Task Tutorial

Key Features:

- Add, update, delete, and display student information.
- Use file handling to store data persistently.
- Implement menu-driven operations for smooth navigation.

Expected Outcome:

A functional application that allows users to add, update, delete, and display student information with persistent data storage for reliable record management.

Completed Successfully!

2

Due Date: 24 Aug 2026

Submitted on track

Bank Management Application

To design and implement a C++ based Bank Management Application that simulates core banking operations using object-oriented programming and file handling.

Watch Task Tutorial

Key Features:

- Include features like deposit, withdrawal, and balance check.
- Use object-oriented programming concepts.
- Store customer data securely using file management.

Expected Outcome:

A secure and functional banking system capable of performing deposits, withdrawals, and balance inquiries while maintaining persistent customer records.

Completed Successfully!

System developed and maintained by Team Emogi (emogi.in)

Thiranex

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3

Due Date: 31 Aug 2026

Submitted on track

Library Management System

To develop a Library Management System that efficiently manages books, members, and borrowing records using structured or object-oriented programming.

Watch Task Tutorial

Key Features:

- Implement structures or classes for book and member details.
- Support book issue and return features.
- Include search functionality by title or author.

Expected Outcome:

A functional application that enables book addition, issue and return processing, and search by title or author for streamlined library operations.

Completed Successfully!

4

Due Date: 07 Sept 2026

Submitted on track

Mini Game Project (Tic Tac Toe / Snake Game)

To create a console-based mini game in C++ that demonstrates core programming concepts such as loops, arrays, and conditional logic.

Watch Task Tutorial

Key Features:

- Use loops, arrays, and condition statements for game logic.
- Display the game board dynamically after each move.
- Add win/loss detection and replay options.

Expected Outcome:

An interactive game with dynamic board display, win/loss detection, and replay functionality, showcasing effective implementation of game logic.

Completed Successfully!

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OFFICIAL CAREER PERK - FREE LOR

Official Letter of Recommendation (LOR)

Earn an authoritative, verifiable Letter of Recommendation signed by our CEO to elevate your resume, placements, and master's applications!

WHY THIS LOR MATTERS:

- CEO & Founder Endorsed: Certifies your technical mastery.
- Placement Edge: Proven credibility in interviews.
- Higher Studies Proof: Accepted for MS/MBA.
- Online Verification: 24/7 instant authentication.

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Claim Your LOR

Download your certificate, post it on LinkedIn celebrating your work, and paste your post link below.

1. Download Certificate

2. Submit LinkedIn Post

System developed and maintained by Team Emogi (emogi.in)

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 4-week Virtual Internship undertaken under the THIRANEX Virtual Internship Program, in the domain of “**C++ Programming**”. The internship was carried out from August 2026 to September 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida.

The primary aim of this internship was to bridge the gap between theoretical knowledge of C++ programming and its practical application in software development. The internship provided hands-on exposure to fundamental and intermediate C++ concepts, including object-oriented programming, functions, data structures, file handling, input validation, and problem-solving techniques. I was encouraged to apply these concepts through practical programming tasks and application-oriented projects.

Throughout the internship, I worked on multiple C++ projects designed to strengthen programming skills and develop a better understanding of software development practices. The projects included applications such as a ‘Student Management System’, ‘Bank Management System’, ‘Library Management System’, and ‘Tic Tac Toe game’, providing practical experience in designing program logic, managing data, implementing user interactions, and handling different input conditions.

The internship followed a structured, week-wise curriculum that combined conceptual learning with hands-on assignments, weekly assessments, project development, and documentation. Each project provided an opportunity to apply programming concepts in a practical environment while improving logical thinking, debugging, and problem-solving abilities. On successful completion of the required activities, including the final assessment test, a Virtual Internship Certificate was awarded by THIRANEX.

4.2 Organization Profile

‘Thiranex – Skill Development & Future Tech’ is a software development and technology learning platform dedicated to fostering talent and innovation across various domains. The organization provides virtual internship programs designed to bridge the gap between academic learning and industry requirements by offering hands-on experience in cutting-edge technologies such as Web Development, App Development, Data Science, Cyber Security, and Python/Java programming. Through real-world project assignments, remote learning frameworks, and structured mentorship, Thiranex helps students and aspiring tech professionals strengthen their technical skill sets, build practical portfolios, and enhance their overall employability.

4.3 Problem Statement

The project aims to develop practical C++-based applications that solve simple real-world problems while demonstrating fundamental programming concepts.

The internship tasks focus on creating:

Student Management System

Bank Management Application

Library Management System

Mini Game Project (Tic Tac Toe/ Snake Game)

4.4 Project Objectives

- To develop practical applications using C++ programming.
- To understand and implement C++ fundamentals.
- To use conditional statements and loops for program control.
- To work with String, OOPS, Functions.
- To implement basic Methods for storing and retrieving information.
- To automate repetitive tasks using C++ scripts.

- To develop a simple rule-based Management System using predefined responses.
- To improve problem-solving and logical-thinking skills.
- To create simple, user-friendly command-line applications.

4.5 Scope of the Project

The scope of this project covers the development of simple and practical C++-based applications designed to demonstrate fundamental programming and problem-solving skills. The project includes the implementation of tasks such as a text-based Management System and Game. It involves the use of C++ programming concepts including variables, conditional statements, loops, functions, strings, user input/output. The developed applications are intended for learning and demonstration purposes and operate primarily in a local or command-line environment.

4.6 Technologies and Tools Used

- Programming Language : C++
- Development : C++ IDE/VS Code /IDLE
- Version Control : Git and GitHub
- Data Structure: OOPS, String, Functions
- User Interaction: input() / print()
- Logic: if-else, loops, functions

4.7 System Architecture

The Student Management System follows a **modular, console-based architecture** developed using C++. The system uses a menu-driven interface to allow users to manage student records efficiently. Data is stored persistently using **file handling**, so records remain available even after the program is closed.

Architecture Components

1. **User Interface Layer**
 - Provides a console-based menu.
 - Allows users to select operations such as Add, Update, Delete, Search, and Display.
 - Accepts user input and displays system results.
2. **Application/Business Logic Layer**

- Processes user requests.
 - Performs operations on student records.
 - Validates student information and manages program flow.
 - Implements functions such as adding, updating, deleting, searching, and displaying records.
3. **Student Data Model**
 - Represents student information using C++ structures/classes.
 - Stores details such as Student ID, Name, Course, Age, Contact, etc.
 - Provides organized access to student records.
 4. **File Handling / Data Storage Layer**
 - Stores student records in files such as `.txt` or `.dat`.
 - Reads existing records when required.
 - Updates and deletes records while maintaining persistent storage.

4.8 Methodology

The Student Management System is developed using a **step-by-step modular methodology** in C++. The development process focuses on efficient student record management, persistent data storage, and simple user interaction.

1. Requirement Analysis

Identify the basic requirements of the system, including adding, updating, deleting, searching, and displaying student records.

2. System Design

Design a **menu-driven console application** with separate functions for each major operation. Define the student data structure/class and file storage mechanism.

3. Implementation

Develop the system using **C++ and Object-Oriented Programming (OOP)** concepts. Implement functions for:

- Add Student
- Update Student
- Delete Student
- Search Student
- Display All Students

4. File Handling

Use C++ file handling to store student records permanently. The system reads data from files when required and writes updated information back to the storage file.

5. Input Validation

Validate user inputs such as Student ID and other required details to reduce invalid or duplicate records and ensure reliable data management.

6. Testing

Test each module individually and then test the complete system to verify that all operations work correctly and data is stored and retrieved properly.

7. Deployment and Usage

After successful testing, the application can be compiled and executed as a **console-based C++ application**, allowing users to manage student records through the menu interface.

4.9 Expected Outcomes

The **Student Management System** successfully provides a simple and reliable way to manage student records through a console-based C++ application. The system demonstrates effective use of **Object-Oriented Programming, file handling, and menu-driven operations**.

Key Outcomes

- Successfully adds, updates, deletes, searches, and displays student records.
- Provides **persistent data storage** using file handling.
- Reduces manual effort in maintaining student information.
- Provides a simple and user-friendly console interface.
- Demonstrates practical implementation of **C++ OOP concepts**.
- Maintains student records efficiently for future retrieval.
- Improves accuracy and organization of student data management.

Final Expected Outcome

A functional Student Management System that efficiently manages student records with persistent file storage, providing reliable and easy-to-use operations for adding, updating, deleting, searching, and displaying student information.

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter and the Internship Completion Certificate issued by Thiranex India , confirming successful completion of the 4-weeks Thiranex Virtual Internship Program in C++ Programming, are attached in Chapter 3 of this report.

THIRANEX VIRTUAL INTERNSHIPS

Official Certificate Ready for Issuance

Dear **Sukrit Kumar**,

Your internship task submissions for **C++ Programming** have been evaluated and **APPROVED**.

Intern ID:	THX-AUG1026-450
Domain:	C++ Programming
Evaluation Status:	✓ Verified & Approved

Your official MSME-registered Internship Completion Certificate is ready for verification. Complete the process below to claim your credential.

5. BIBLIOGRAPHY/REFERENCES• Bjarne Stroustrup, *The C++ Programming Language*, 4th Edition, Addison-Wesley Professional.

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2. [C++ Reference – File Input/Output \(fstream\)](#)
3. [C++ Reference – Classes and Objects](#)
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